

# Probability

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- **Experimental Probability:** The probability obtained from the result of an experiment when we actually perform the experiment is called experimental (or empirical) probability.
- **Theoretical Probability:** The probability we find through the theoretical approach without actually performing the experiment is called theoretical probability.
- The theoretical probability (or classical probability) of an event E, is denoted by  $P(E)$  and is defined as

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes of the experiment}}$$

- Experimental probability may or may not be equal to the theoretical probability.
- Formula of theoretical probability can be used to find the probabilities of various events.

**Example 1:** A dice is thrown once. What is the probability of getting 1 on the dice?

**Solution:** When a dice is thrown once, the possible outcomes are 1, 2, 3, 4, 5, 6. Let A be the event of getting 1 on the dice.

$$\therefore P(A) = \frac{\text{Number of outcomes favourable to A}}{\text{Number of all possible outcomes}} = \frac{1}{6}$$

**Example 2:** A box contains 3 white, 5 green, and 6 red balls. A ball is drawn at random out of the box. Find the probability of drawing a red ball.

**Solution:** Total number of balls =  $3 + 5 + 6 = 14$

Therefore, total number of possible outcomes = 14

Let E be the event of drawing a red ball.

Number of outcomes favourable to E = 6

$$\therefore P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes}} = \frac{6}{14} = \frac{3}{7}$$

**Example 3:** A card is drawn at random from a deck of 52 cards. Find the probability that the card drawn is,

- - A black card
  - An ace
  - Neither a black card nor an ace

**Solution:**

Since there are 52 cards in a deck, the number of all possible outcomes is 52.

- - Number of black cards in the deck = 26

Therefore, the probability that the drawn card is black  $= \frac{26}{52} = \frac{1}{2}$

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- Number of aces in the deck = 4

Therefore, the probability that the drawn card is an ace  $= \frac{4}{52} = \frac{1}{13}$

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- Number of cards which are neither black nor an ace =  $26 - 2 = 24$

Therefore, the probability that the drawn card is neither black nor an ace  $= \frac{24}{52} = \frac{6}{13}$

- **Complementary events**

For an event E such that  $0 \leq P(E) \leq 1$  of an experiment, the event  $\bar{E}$  represents 'not E', which is called the complement of the event E.

We say, E and  $\bar{E}$  are **complementary** events.

$$P(E) + P(\bar{E}) = 1$$

$$\Rightarrow P(\bar{E}) = 1 - P(E)$$

**Example:**

A pair of dice is thrown once. Find the probability of getting a different number on each die.

**Solution:**

When a pair of dice is thrown, the possible outcomes of the experiment can be listed as:

|   | 1      | 2      | 3      | 4      | 5      | 6      |
|---|--------|--------|--------|--------|--------|--------|
| 1 | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (1, 5) | (1, 6) |
| 2 | (2, 1) | (2, 2) | (2, 3) | (2, 4) | (2, 5) | (2, 6) |
| 3 | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
| 4 | (4, 1) | (4, 2) | (4, 3) | (4, 4) | (4, 5) | (4, 6) |
| 5 | (5, 1) | (5, 2) | (5, 3) | (5, 4) | (5, 5) | (5, 6) |
| 6 | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |

The number of all possible outcomes =  $6 \times 6 = 36$

Let E be the event of getting the same number on each die.

Then,  $\bar{E}$  is the event of getting different numbers on each die.

Now, the number of outcomes favourable to E is 6.

$$\therefore P(\bar{E}) = 1 - P(E) = 1 - \frac{6}{36} = \frac{5}{6}$$

Thus, the required probability is  $\frac{5}{6}$ .